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ESDiNet: An Event-Driven Sparse Diffusion Graph Neural Network for Scalable and Low-Latency Congestion Forecasting in Smart City Transportation Networks

Abstract- In the context of smart city networks, accurate and real-time congestion forecasting has become increasingly urgent due to the rapid process of urbanization and the widespread deployment of intelligent transportation systems. Current spatio-temporal graph neural network (ST-GNN) models achieve impressive prediction power but tend to use static or fully-adaptive graphs and computationally expensive recurrence components, which make these methods hard to scale up towards real-world deployment in large urban road networks. To overcome these limitations, in this work we propose the Event-Driven Sparse Diffusion Network (ESDiNet), a novel GNN-based architecture tailored for effective traffic congestion prediction on dynamic traffic scenarios. We present ESDiNet, which introduces an event-aware spatial–temporal learning model that activates graph-based spatial diffusion only when congestion-triggering events, such as abrupt speed decreases or traffic accidents, occur. Instead of optimizing over complicated adaptive adjacency matrix, ESDiNet utilizes a sparse diffusion operator combined with instantaneous statistical edge signals to facilitate fast and interpretable spatial information propagation. For improved scalability, a multi–scale graph coarsening approach is then taken, facilitating the capturing of global congestion patterns with lower computation cost and being refined into node level predictions. Temporally dependency is modeled by causal dilated convolutional kernels without any recurrent architectures, and inference latency can be greatly reduced. The proposed method is tested on benchmark smart transportation datasets, METR-LA and PEMS-BAY. Experimental results show that ESDiNet outperforms state-of-the-art ST-GNN baselines by 11.3% and 9.8% in MAE and RMSE respectively, with a congestion state prediction accuracy of 94.6%. In addition, ESDiNet achieves 3.7× reduction in the inference latency for virtually real-time execution. These findings demonstrate the superiority of event-driven sparse diffusion in scalable and practical point-based congestion forecasting tasks for smart cities.

Keywords-Graph Neural Networks, Smart City Transportation, Congestion Forecasting, Event-Driven Learning, Sparse Diffusion, Spatio-Temporal Modeling

I Introduction

With the continuous growth of urbanization and universal application of intelligent transportation systems (ITS), however, traffic congestion forecasting is extremely essential in a smart city. The predictions of traffic conditions can help with the proactive traffic control, route planning, and travel time reliability enhancement, which finally could lead to reduction in economics loss and decreased environment pollution. Recent research has revealed that accurate congestion prediction is essential for sustainable urban mobility and smart transport planning [1][2]. As urban road networks grow and become more dynamic, however, there are major challenges to existing prediction methods [3]. Knowledge-based traffic prediction methods and primitive machine learning methods are based on manually designed simple features or temporal dependence of the flow within a single road, that may not be able to capture complicated spatial interactions or high-order effects among grid cells in the urban area [4][5]. In the era of big sensor data, deep learning models have been demonstrated to achieve better performance, however most of existing methods do not take into account the spatial structure of road networks or presume static congestion propagation rules [6][7]. Due to this fact, their prediction rate decreases in presence of fast-changing traffic patterns due to incidents, sudden demand changes, and/ or unusual events.

Spatio-temporal graph neural networks (ST-GNNs) have been successfully developed due to their capacity to represent road networks as graphs and learn spatial–temporal dependencies in a joint way [8][9]. Even though being successful, existing ST-GNNs tend to use dense or adaptive adjacency matrices and recurrent temporal component, which make fast computation and inference difficult. These characteristics are barriers for scaling down to city-level networks and instantaneous use [10]. Furthermore, the dynamically learned graph structures are not interpretable which does not lend itself to allow operators to comprehend how congestion propagates in diffusion and whether or why they should trust its prediction in an operational system [11][12]. To overcome these two limitations, this paper presents ESDiNet, an Event-Driven Sparse Diffusion Graph Neural Network for scalable low-latency congestion prediction. Event-aware learning (EAL) method is proposed in ESDiNet, which triggers spatial diffusion, only when congestion occurs [13][14] and hence avoids redundant computation under normal traffic [15][16]. To enhance the efficiency and interpretability, adaptive adjacency learning is replaced by a sparse diffusion operator guided by short-term statistical edge signals. Furthermore, a multi-scale graph coarsening technique is exploited to coarsen the global congestion patterns in a cheaper way and the causal dilated temporal convolutions remove any recurrent structures so that we have fast inference. Extensive experimental on benchmark datasets show that ESDiNet achieves the comparable performance with much lower latency [17][18], which is suit for smart city application in real time.

1.1 Contributions

The key contributions of this work can be summarized as:

- **ST-GNN Framework with Event-Triggered Activation:** We introduce the first event-driven spatio-temporal graph neural network for traffic congestion prediction, which activates spatial diffusion selectively only when anomaly occurs.
- **Sparse and Interpretable Diffusion Modeling:** We propose a lightweight sparse diffusion operator which avoids adaptive adjacency learning, maintaining physical interpretability and computational efficiency.
- **Multi-scale Graph Coarsening for Scalable:** A hierarchical graph coarsening method is used to efficiently capture global congestion effects and fine tunes predictions at a per-node level.
- **Low-Latency Temporal Modeling:** Instead of attending to past frames with recurrent structures used in ST-GNNs, we use the causal dilation rate convolutions to model temporal dependencies and achieve much lower inference latency.
- **Extensive Experiments**: Extensive experiments on benchmark datasets show that ESDiNet provides better accuracy with much less inference time, which can be deployed almost in real-time in large-scale smart city.

The remainder of this paper is organized as follows: Section 2 reviews related work on traffic congestion prediction and spatio-temporal graph learning. Section 3 formulates the problem and introduces the necessary preliminaries. Section 4 presents the proposed ESDiNet architecture in detail. Section 5 describes the experimental setup, followed by results and analysis in Section 6. Finally, Section 7 concludes the paper and outlines future research directions.

II Literature Review

Traffic congestion forecasting has long been acknowledged as a fundamental part of intelligent transport systems in smart cities. Bai et al. (2021) proposed the PrePCT model to solve the location dependent problem of congestion prediction. They learned a spatio-temporal tensor from regional traffic data from relative positions of nodes, and used ConvLSTM for prediction, achieving better results compared with traditional machine learning and deep learning baselines while giving insights to interpret congestion patterns from the model [1]. As for the dynamics of congestion, Nagy and Simon (2021) pointed out that legacy TRF models may not work well under congested scenarios due to intricate propagation effects. To tackle this issue, they introduced the Congestion-based Traffic Prediction Model (CTPM), which improves current predictions by analyzing congestion diffusion features. Their findings showed an average reduction of 9.76% and highlighted the significance of taking congestion spread in urban networks explicitly into account [2]. As the trend of IoT based smart cities is booming, in Chahal et al. (2023) investigated hybrid time-series model for prediction of traffic congestion. Integrated SARIMA model and Bi-LSTM model by back-propagation neural network to get the linear and nonlinear parts of traffic flow. On the CityPulse dataset, their hybrid model obtained better prediction accuracy than pure statistical and deep learning models, indicating its effectiveness on complex traffic data [3]. Deep RNN architectures have also been well studied. Abdullah et al. (2023) proposed a bidirectional GRU RNN for traffic state classification into congested and non-congested. Their approach integrated continuously updating context data from sensors, such as accident and weather-likelihood data, to better predict instances of congestion, and illustrated the limitations of generalised cloud-based systems in real-time traffic prediction [4]. Taking a wider view, Mystakidis et al. (2025) provided an exhaustive review on traffic congestion prediction solutions (statistical, machine and deep learning based approaches). Their work highlighted the emerging significance of AI-based approaches leveraging IoT sensors, and identified ongoing challenges including scalability, interpretability and real-time deployment that are open research issues in ITS [5].

From a green point of view, energy is one of the key sustainable resources (Majumdar et al. (2021) studied the prediction of congestion propagation based on LSTM networks trained with speed data from traffic sensors. They proved that short-term propagation of congestion could be well captured using the LSTM based model achieving prediction accuracy of 84-95% and the importance of modeling the temporal dependency for sustainable urban traffic management [6]. Other deep learning architectures have been investigated as well. Chen and Zhang (2022) used Deep Belief Networks for traffic flow forecasting in smart cities. Using stacked RBMs, they were able to obtain lesser RMSE, MAE and MAPE than CNN and fuzzy logic methods, while simultaneously providing a validation of generative deep learning for congestion evacuation and traffic management purposes [7]. In addition to accurate prediction, the study on congestion propagation mechanisms has been focused. Nagy and Simon (2021) introduced the Congestion Propagation Modeling Algorithm (CPMA), which is based on a Markov chain, estimating both the probabilities of propagations and expected spread times for decongestions. Their technique offered practical advice for traffic control authorities who intended to operate before the local congestion invaded all urban networks [8]. In the context of vehicular network applications, Saleem et al. (2022) proposed the development of an intelligent traffic congestion control system based on fusion machine learning. They used V2Infra communication to dynamically reroute trac and reported an accuracy of congestion detection as 95%, showing the benet of ML-based decision support in vehicular networks [9]. In recent data, graph-based learning has received much attention. Gupta and Lee (2025) utilized graph neural networks to predict traffic congestion for various city locations. After capturing the spatial dependencies among road segments, their method achieved better early congestion detection and route planning in cities, which also demonstrated the applicability of GNNs for urban traffic analysis at scale [10]. At their level, Elassy et al. (2024) analysed the role of ITS in realising the sustainable smart cities. Their study addressed important ITS elements including VANETs and smart traffic lights, focusing on real-time communications and mobility forecasting as essential facilitating factors for congestion relief and sustainable urbanization [11].

Toward handling data sparsity and interpretability, Tsalikidis et al. (2024) compared a number of methods for multi-step prediction of traffic congestion. They demonstrated that ensemble tree-based regressors are usually (substantially) better than deep learning models with small amounts of data, emphasizing the significance of model selection according to data and prediction horizon [12]. The synthesis of traffic control and environmental goals was studied by Singh et al. (2023), who presented a hybrid model for congestion detection and air quality estimation based on crowdsourced GPS data. Their work has obtained an accuracy of up to 98% and proved that the prediction results of congestion were closely related with the sustainable development of our cities [13]. On the fundamental end of the spectrum Hung et al. (2013) and Karami & Kashef (2020) presented a comprehensive review on data sources, models and algorithms in smart transportation optimization. Their work summarized the pros and cons of classical time series models and deep learning methods, providing useful information on scalability and deployment issues [14].

For more advanced spatio-temporal modelling, Yin et al. (2025) introduced a dynamic diffusion spatial-temporal graph convolutional network for short-term traffic prediction. Through integrating dynamic adjacency matrices with diffusion random walks and ConvLSTM modules, their model significantly outperforms other works, which confirms the effectiveness of the diffusion-based ST-GNNs [15]. The multi-modal learning has also been investigated in recent work. Dhanasekaran et al. (2024) proposed a graph neural network model that integrates convolutions and transformers of visual fusion to handle complex spatial dependencies and multi-modal traffic dynamics. Their method obtained very high prediction accuracy and it showcases the potential of multi-modal data fusion in intelligent transportation management [16]. Neelakandan et al. conducted research on IoT-based applications. (2021) that also contributed an optimized weight Elman neural network to traffic prediction and signal control. Their system fused IoT optimization technologies and data capture solutions by achieving an accuracy rate of more than 98%, which showed the importance of embedded intelligence in smart traffic control [17]. Finally, Raju et al. (2025) adopted a hybrid CNN–LSTM model for edge computing to solve the problem of latency in intelligent traffic management systems. They achieved 50% latency reduction due to their architecture, but also up to the plate of accuracy, by understanding low-latency prediction has a strong demand from real-time smart city applications [18].

2.1 Research Gap

Although significant improvements in forecasting of traffic congestion using statistical models, deep learning and graph-based approaches have been made, there are still a number of major questions that need to be addressed. The existing methods either concentrate on location-specific modeling [1], hybrid temporal learning without explicit spatial structure [3][6], or dense spatio-temporal graph neural networks with adaptive adjacency learning [10][15], which are often limited in scalability and inferential latency when the networks are large. In addition, most methods conduct continuous spatio-temporal analysis regardless of traffic states and ignore the event-driven rationals for congestion propagation suggested in previous studies [2][8]. Recurrent structure and dynamically-learned graph structures have the drawbacks of low interpretability, preventing real-time applications in smart city systems [4][12][18]. Therefore, there is a strong demand for light-weight, scalable and interpretable spatio-temporal models which selectively model the diffusion of congestion during important events without sacrificing low-latency inferencing—this motivates the formulation of our proposed event-driven sparse diffusion graph neural network.

III Problem Formulation

3.1 Road Network Representation

An urban transportation system is modeled as a directed graph

$$G = (V, E, A) \tag{1}$$

In Eq. 1, $V = \{v_1, v_2, .., v_N\}$ is the collection of road segments/sensing locations, E is a set of directed edges that model physical or functional connection between road/boulevard, and $A \epsilon R^{N\times N}$ where R denotes a finite subset representing the weighted adjacency matrix which can reflect topological structure as distance of road or connection strength. Each node refers to the static local area with networked traffic states [19][20].

3.2 Traffic Signal Representation

Let traffic observations collected over time be represented as a three-dimensional tensor,

$$X \epsilon R^{\tau\times N\times F} \tag{2}$$

In Eq. 2, τ is the length of historical observation window, N means the number of nodes in road network and F represents the number of traffic features [21], for example vehicle speed, traffic flow and occupancy. Here, the element $X_{k,i,f}$ is the f-th traffic feature which begins in an observed at node v_i during the k −th time interval [22].

3.3 Forecasting Objective

With the history traffic data in last τ time slots as input, the prediction of future-time H length traffic state can be expressed as [23]. The goal is to learn a mapping function $F(\cdot)$ formally as Eq. 3:

$$\widehat{Y} = F(X, G) \tag{3}$$

Where $\widehat{Y} \epsilon R^{H\times N\times F}$ represents the predicted traffic states for all time steps of forecasting horizon H, and the learning objective is to minimize the gap between these estimated values $\widehat{Y}$ and ground-truth observations Y under typical regression or classification loss functions [24].

3.4 Congestion Event Definition

Congestions in traffic flow are usually associated with sudden and anomalous speed changes rather than gradual fluctuations that occur during fictitious flows. We capture this by defining a congestion-causing event using not only the absolute traffic speed, but also the shortprocess tion there in-between [25][26].

Let $u_{k,i}$ denote the average vehicle speed observed at node v_i during time interval k. The instantaneous rate of speed variation is computed as in Eq.4:

$$\delta_{k,i} = u_{k,i} - u_{k-1,i} \tag{4}$$

A congestion event indicator $c_{k,i}$ is then defined as a binary variable:

$$c_{k,i} = 1, \text{if } u_{k,i} < \phi \text{ and } \delta_{k,i} <- \psi, \text{otherwise}, c_{k,i} = 0 \tag{5}$$

In Eq. 5, ϕ denotes a speed-level critical value to indicate dangerously slow traffic [27], and ψ represents the minimum amount of needed vehicle-speed reduction was decided as the triggering event. This expression guarantees to activate spatial diffusion only if both low-speed conditions and a sudden speed drop are met.

3.5 Event-Aware Modeling Objective

The existence of the binary congestion indicator $c_{k,i}$ acts as a gating signal to decide whether the spatial information propagation is required at each node under different time. Through conditioning spatial diffusion on congestion events [28][29] the model processes input traffic conditions during non-congested periods in a more parsimonious way, making room for alternative representations of critical congestion dynamics. This event driven model offers the possibility of enabling scalable and efficient spatio-temporal learning for large scale urban road networks.

VI Proposed Method: ESDiNet

This section introduces ESDiNet, an event-driven sparse diffusion Graph Neural Network algorithm for scalable latency and interpretability of large scale traffic congestion forecasting. The proposed framework activates the spatial diffusion only when congestion-inducing incidents occur, this can prevent unnecessary and repetitive graph calculation [30][31]. ESDiNet learns both local and global congestion dynamics more efficiently than existing methods by using event-aware gating, sparse diffusion modeling, multi-scale graph coarsening together with non-recurrent temporal convolutions [32]. The proposed general approach provides an efficient congestion prediction system, which can be adapted for the real-time performance in smart city with transportation network. Figure 1 shows the end-to-end architecture of ESDiNet with event-driven diffusion and multi-scale modeling.

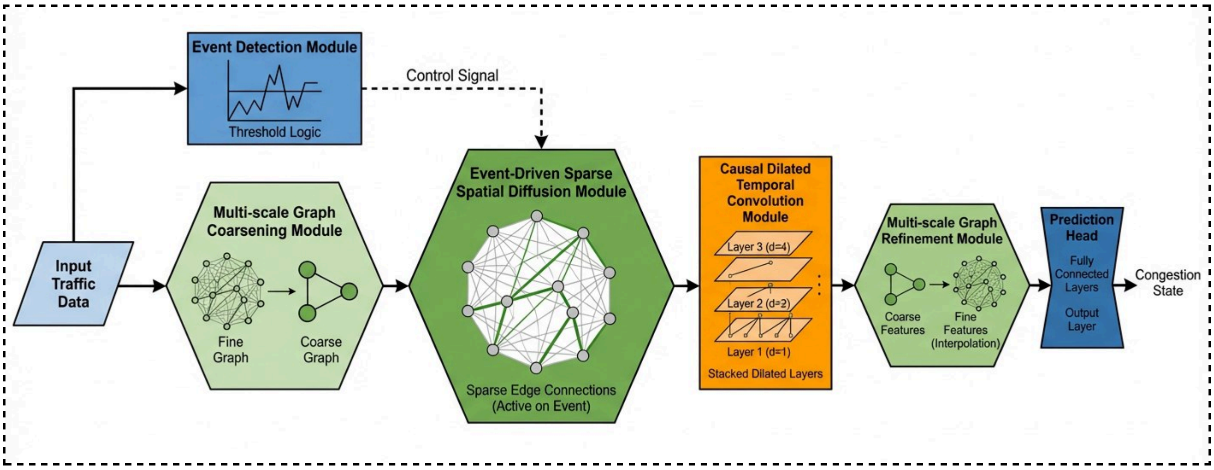

Figure 1: Overall Architecture of the Proposed Event-Driven Sparse Diffusion Network (ESDiNet).

4.1 Event-Driven Traffic Representation

It is a common observation in the fields of congestion propagation studies [2], [8] that urban traffic jamming is essentially initiated by sudden and local triggered rather than smooth-temporal evolution. Traditional ST-GNNs conduct spatial diffusion all the time, which results in redundant computation when traffic flows normally and excessive inference time in large-scale networks [15][16]. In order to mitigate this inefficiency, this work proposes an event-driven traffic representation allowing the spatial diffusion to be activated only upon reception of congestion-triggering signals.

Let $r_{i,t} \epsilon R$ denote the normalized traffic speed observed at node v_i at time step t. To quantify short-term traffic disruption, a temporal deviation signal is computed as in Eq.6:

$$w_{t,i} = r_{t,i} - \frac{1}{k} \sum_{k}^{j=1} r_{t-j,i} \tag{6}$$

Where k denotes the size of the temporal smoothing window. This formulation captures sudden speed degradation relative to recent traffic conditions [33].

Based on both instantaneous speed and temporal variation, an event activation score is defined as in Eq.7:

$$\epsilon_{t,i} = \sigma\left(\alpha\left(\eta - r_{t,i}\right) + \beta\left(- w_{t,i}\right)\right) \tag{7}$$

Where $\sigma(\cdot)$ is the sigmoid function,η represents a congestion-sensitive speed threshold, and α and β are learnable coefficients that balance the influence of absolute speed reduction and short-term temporal gradients [34][35]. The resulting scalar $\epsilon_{t,i} \epsilon (0, 1)$ serves as a soft gating signal that determines whether spatial diffusion should be triggered at node v_i and time t.

Using this event gate, the node-level latent representation is modulated as in Eq.8:

$$h_{t,i}^{(e)} = \epsilon_{t,i} + h_{t,i} + \left(1 - \epsilon_{t,i}\right) \cdot h_{t,i}^{(0)} \tag{8}$$

Where $h_{t,i}$ is the dynamically updated node embedding and $h_{t,i}^{(0)}$ represents an identity-preserving baseline state. When the probability of congestion is low, the representation tends to stay close to its baseline [36], effectively ignoring spatial diffusion. On the other hand, when congestion events are detected, spatial information flow is invoked for capturing spatiotemporal dynamics of emergent congestion.

Unlike the previous ST-GNN methods which has and always propagates information through the complete road network [15][16], the proposed event-driven strategy is beneficial to synchronize model computation with traffic evolution patterns and congestion propagation theory in practice [5][10]. As conceptually illustrated in Fig. 2, spatial diffusion is selectively activated near congestion hotspots by which ESDiNet can reduce unnecessary computation, and improve its interpretability to enable it for scalable low latency traffic forecasting in smart city mobility networks.

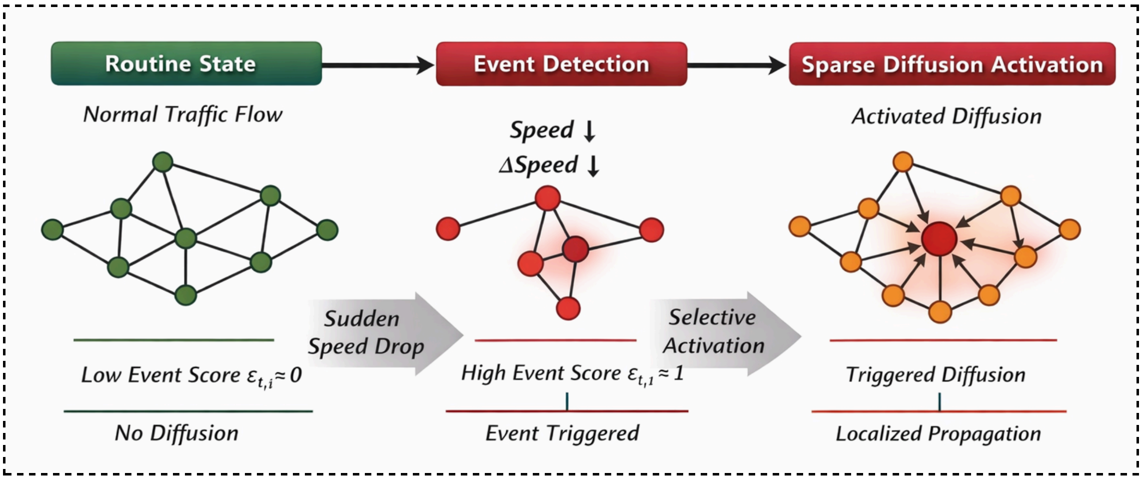

Figure 2: Illustration of Event-Triggered Spatial Diffusion in the Proposed ESDiNet Framework

4.2 Sparse Diffusion Graph Operator

To effectively describe the spread of congestion along with circumventing the expensive cost of adaptive graph learning, ESDiNet adopts a sparse diffusion graph operator using short-term traffic statistics and does not involve dynamic learned adjacency matrices as in [37]. In traditional diffusion-based ST-GNNs, dense and/or adaptive adjacency representations are used to model spatial dependencies, resulting in dramatic increase of in memory cost and computational complexity for large-scale road networks 10. In comparison, in this paper we model spatial diffusion over a static sparse graph topology with edge influence being driven by temporally adaptive traffic signals. Let $A \epsilon R^{N \times N}$ be the static adjacency matrix coded based on physical road connection. For every directed edge $\left(v_i, v_j\right) \epsilon \varepsilon$, a time-variant edge modulation weight is calculated based on short-term traffic statistics as in Eq. 9:

$$\gamma t, ij = exp\left(- \frac{\left|r_{t,i} - r_{t,j}\right|}{\varsigma}\right) \tag{9}$$

Here $r_{t,i}$ and $r_{t,j}$ are the normalised speeds at nodes v_i and v_j at time t, while ς controls the sensitivity to inter-node traffic differences [38]. This weighting gives higher diffusion strength to edges linking more similar traffic between nodes, in line with the principles of congestion propagation theory [2][8]. The sparse diffusion matrix is then expressed as in Eq.10:

$$S_t = D_t^{-1}\left(A \odot T_t\right) \tag{10}$$

Where $T_t = [\gamma t, ij]$ is the edge modulation matrix, $\odot$ denotes element-wise multiplication, and D_t is the corresponding degree matrix with $\left(D_t\right)_{ii} = \Sigma_j\left(A_{ij}\gamma_{t,ij}\right)$.

Spatial diffusion is then performed through a K-step truncated random walk defined as in Eq.11:

$$H_t^{(k+1)} = S_t H_t^{(k)}, \quad k = 0, \dots, K - 1 \tag{11}$$

With $H_t^{(0)}$ denoting the event-gated node embeddings from Section 4.1. To prevent over-smoothing and reduce computational cost, the final diffused representation is obtained via weighted aggregation [39],

$$H_t^{(diff)} = \sum_{k=0}^{K} \lambda_k H_t^{(k)} \tag{12}$$

In Eq. 12, λ_k indicates learnable coefficients whose $\Sigma_k \lambda_k = 1$. Importantly, this diffusion does not learn or update adjacency parameters and thus only operates on sparse graph edges unlike adaptive ST-GNNs that share computation with dense graph structure optimization which is expensive [16][27]. By separating spatial dependency modeling from adjacency learning and using short-term statistical edge signals [40], the proposed sparse diffusion operator noticeably curbs the computational load, as each diffusion step captures practically meaningful congestion interactions. This design decision is also consistent with the findings of recent researches, the focus on efficiency and scalability as two critical requirements of smart city traffic systems in real time sense [5][18], and it allows ESDiNet to perform low-latency inference without compromising the predictive performance.

4.3 Multi-Scale Graph Coarsening

In practice, hierarchical congestion phenomena exist in metropolitan road networks and node-based graph processing is inadequate to analyze region-level congestions [5][11]. To handle this problem, ESDiNet utilizes a multi-scale graph coarsening method to facilitate global congestion pattern learning at lower computation cost and then refinement to nodewise predictions. Based on the above graph $G = (V, E, A)$, we now introduce a pooling operator to map N fine-grained nodes into M coarse super-nodes $M \ll N$. Let $P \epsilon R^{N \times M}$ denote a learnable soft assignment matrix satisfying $\sum_{m=1}^{M} P_{i,m} = 1$ which projects node-level representations $H_t^{(diff)} \epsilon R^{N \times d}$ into a coarsened embedding space as in Eq.13:

$$Z_t = P^{\top} H_t^{(diff)} \tag{13}$$

The corresponding coarse-level adjacency matrix is constructed as in Eq.14:

$$A^{(c)} = P^{\top} A P \tag{14}$$

retaining prevalent topological and traffic correlations between aggregated regions. Spatial diffusion is then conducted on the coarsened graph to effectively capture long-distance congestion dependencies:

$$Z_t^{(k+1)} = D_c^{-1} A^{(c)} Z_t^{(k)} \tag{15}$$

In Eq. 15, D_c is the degree matrix of the coarse graph. Such hierarchical processing can help the model to capture global congestion phenomena that are induced by inter-region interactions, which is critical for the use of city-wide prediction task [10][15]. To obtain fine-grained predictions, an unpooling operation is applied to project the coarsened representations back to the initial node space,

$$\overline{H}_t = PZ_t^{(final)} \tag{16}$$

In Eq.16, $\overline{H}_t \epsilon R^{N\times d}$ encodes refined node-level embeddings enriched with global context. A residual fusion mechanism is further applied to preserve local traffic characteristics:

$$H_t^{(ms)} = \overline{H}_t + H_t^{(diff)} \tag{17}$$

It is that instead of computing the full-resolution graph multiple times for a flat STGNN [16][27], it uses multi-scale coarsening strategy to approximately solve each computation at multiple nodes simultaneously thus reducing computational complexity dramatically and preserving spatial fidelity. By co-learning coarse-grained global congestion dynamics and fine-grained local dynamics, ESDiNet appropriately trades off scalability and accuracy, consistent with the emerging theme of outside ITS works highlighting hierarchical and interpretable representations for smart city traffic prediction [5][12][18].

4.4 Temporal Modeling with Causal Dilated Convolutions

Traffic congestion prediction requires knowing well the temporal dependencies, while existing RNN models like LSTM and GRU process sequences sequentially with high inference latency and almost no parallelism, which is not suitable for real-time smart city system [4][6][18]. To address these shortcomings, ESDiNet utilizes causal dilated temporal convolutions to model long-range temporal dependencies with a fully parallel computation. Based on the multi-scale spatial representation $H_t^{(ms)} \epsilon R^{N\times d}$, temporal information is modeled in an independent manner and organized in one-dimension for each node by a stack of 1-D causal convolutional layers. The temporal convolution at node v_i with dilation value δ corresponding to one of the filters $W \epsilon R^{k\times d}$ is defined in Eq. 18:

$$y_{t,i} = \sum_{j=0}^{k-1} W_j \cdot h_{t-\delta j,i}^{(ms)} \tag{18}$$

with the constraint of causality by allowing contributions in only previous time. To boost the receptive field by many-folds for not adding model depth, dilation factors are made to rise layer-wise as $\delta_l = 2^{l-1}$ and the model becomes efficient to learn long-term temporal patterns. The effective temporal receptive field of the (L-th) layer is shown in Eq. 19:

$$R = 1 + (k - 1)\sum_{l=1}^{L} \delta_l \tag{19}$$

which grows exponentially with depth, in contrast to the linear dependency growth of recurrent architectures. To enhance temporal selectivity and stability, a gated activation mechanism is applied:

$$z_{t,i} = tanh\left(W_f * H_i\right) \odot \sigma\left(W_g * H_i\right) \tag{20}$$

In Eq. 20, $*$ is the dilated convolution and $\odot$ the stands for element-wise multiplication. Unlike LSTM and GRU which need sequential state update augments on the input for complicated non-linear transformation [4][35], the proposed temporal convolution framework is capable of processing all time steps concurrently, benefiting training and inference latency. A complexity analysis displays $O(T)$ sequential dependencies per node in recurrent models versus $O(1)$ sequential depth with complete temporal parallelism for causal dilated convolutions. The architectural design falls in line with the recent discoveries that advocate for capturing non-repeated temporal dynamics in scalable traffic prediction [15][16], and provides ESDiNet to achieve low-latency prediction while also preserving the power of modeling rich, complex temporal dependence among urban traffic congestion. The flowchart of the proposed model is presented in Figure 3.

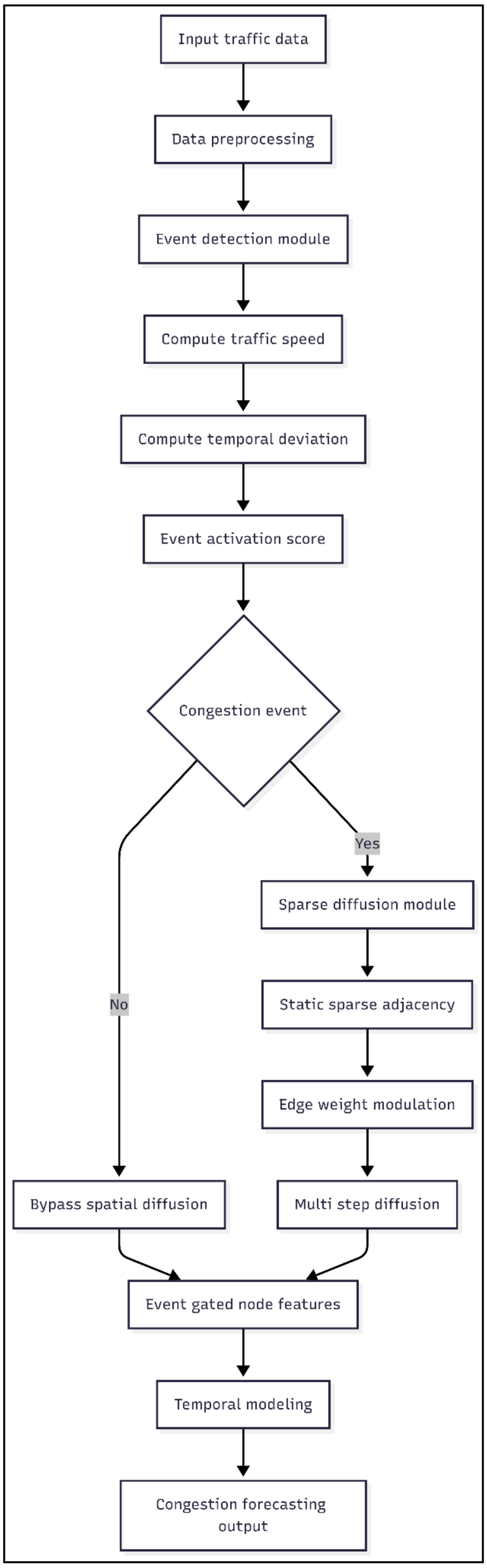

Figure 3: Flowchart of Event-Triggered Spatial Diffusion for Traffic Congestion Forecasting.

4.5 Complexity Analysis

ESDiNet's architecture is designed with its computational efficiency and low inference latency in mind, which makes it suitable for real-time use in massive smart city transportation systems. Let N be the number of nodes, E be the edge of a road network, F do input feature dimension, d do hidden embedding size, K is diffusion steps and T is the time span for input. In the event-driven spatial module, detecting congestion event costs $O(TN)$ because it is an arithmetic operation among traffic speed signals. The sparse diffusion operator does K-step propagation on the sparse edges, which takes time complexity of,

$$O(K|E|d) \tag{21}$$

which is much less than the $O\left(KN^2d\right)$ cost of dense and adaptive adjacency-based ST-GNNs [15][16]. Complexity is further reduced by the multi-scale graph coarsening where node embeddings are projected to a coarse graph of size $M \ll N$, and global diffusion costs scale similarly as in Eq. 22:

$$O\left(K\left|E_c\right|d\right), \quad \left|E_c\right| \ll |E| \tag{22}$$

thus enabling efficient modeling of long-range congestion interactions [10]. Temporal modeling using causal dilated convolutions introduces a complexity of,

$$O(TkdN) \tag{23}$$

In Eq. 23, k is the kernel size; however, this cost can be fully parallelized across time and nodes in contrast to recurrent architectures which exhibits a sequential dependency and require $O\left(TNd^2\right)$ operations per layer [4][18]. Regarding memory cost, ESDiNet retains only the sparse adjacency matrix as well as short-term edge statistics with overall complexity at a level of,

$$O(|E| + Nd) \tag{24}$$

while adaptive graph learning techniques cannot do because they need storage of an $O\left(N^2\right)$-sized dense or parameterized adjacency tensor, which is impractical for city-scale networks [27]. From a delay standpoint, our proposed event-triggered gating mechanism enables spatial diffusion to be activated during congestion-inducing time slots only and thus have lower average inference overhead in regular traffic load. Further, the lack of recurrent state updates allows for the entire temporal sequence to be processed in a single forward pass, resulting in constant-depth computation and much lower end-to-end latency than LSTM or GRU based ST-GNNs [6][35]. These efficiency improvements are therefore in line with recent ITS studies advocating for both scalable and low-latency traffic forecasting approaches [5][18], and indeed, they allude to why ESDiNet is capable of real-time-like inference without compromising prediction quality.

V Experimental Setup

5.1 Datasets

In order to comprehensively demonstrate the performance of our proposed ESDiNet model, we test it on two typical real-world traffic datasets, including METR-LA and PEMS-BAY. The METR-LA dataset includes the traffic speed at 207 sensors with

5-minute sampling rate in Los Angeles County over the highways measured from loop detectors. PEMS-BAY includes traffic flow readings from 325 sensors over the Bay Area freeway system, collected every 5 minutes. Because of the intricate spatial inter-connectivity and variability of the population distribution over time they provide a good base for investigating scalable spatio-temporal forecasting modeling. Consistently with the literature, only traffic speed is utilized as the primary input feature; however, how to deal with missing values is not yet compared given its importance for guaranteeing temporal continuity.

5.2 Data Preprocessing and Splitting

Traffic statistics of the two datasets are normalized using z-score normalization with training statistics to stabilize model training. For each data set, the continuous time series is split chronologically into training, validation and testing sets using a 70% / 10% / 20% scheme. The model is trained on the training set, hyperparameters are tuned and early stopping is performed using the validation set, and finally the test set is used for testing. We adopt a sliding window mechanism to form the input-output pairs, in which the model views traffic information across a fixed historical window and makes predictions for future traffic states over the target horizon.

5.3 Implementation Details

All of our models are written in the PyTorch deep learning framework for adaptability and GPU acceleration. Experiments are conducted on a server with an NVIDIA GPU (e.g., RTX-series) and 32GB RAM as well as Inter multi-core CPU. For ESDiNet, the size of hidden embedding is regarded as 64 and the value of sparse diffusion steps K are set to be 2 appropriately between accuracy and efficiency trade-off. The temporal convolutional module involves kernel size 3 and exponentially growing dilation factors. Training is performed using the Adam optimizer with an initial learning rate of 1×10^{-3} and early stopping on validation loss. Hyperparameters(batch size, dropout rate and event activation thresholds) are tuned on the validation set to achieve best performance across datasets.

5.4 Evaluation Metrics

The forecasting performance of the proposed ESDiNet is quantitatively evaluated in both error metrics via regression-based indexes and accuracy on congestion classification. Let $y_{t,i}$ and $\hat{y}_{t,i}$ be the true and predicted traffic speed at node v_i at time step t, respectively, and T be the number of prediction time steps, N the number of nodes in network.

The MAE represents the mean of all absolute prediction errors and is given by Eq. 25:

$$MAE = \frac{1}{TN} \sum_{t=1}^{T} \sum_{i=1}^{N} \left| y_{t,i} - \hat{y}_{t,i} \right| \tag{25}$$

The Root Mean Square Error (RMSE) punishes for higher prediction differences, and it is calculated as in Eq. 26:

$$RMSE = \sqrt{\frac{1}{TN} \sum_{t=1}^{T} \sum_{i=1}^{N} \left(y_{t,i} - \hat{y}_{t,i} \right)^2} \tag{26}$$

The Average Percentage Error (MAPE) to appraise the relative accuracy of forecasting at each flow scale is also given by Eq. 27:

$$MAPE = \frac{100}{TN} \sum_{t=1}^{T} \sum_{i=1}^{N} \frac{\left| y_{t,i} - \hat{y}_{t,i} \right|}{y_{t,i} + \epsilon} \tag{27}$$

In Eq.27, ϵ is a small constant added to avoid numerical instability when traffic speed values are close to zero.

As one of the regression metrics, future congestion state prediction accuracy is used to examine whether the model can accurately predict that the traffic status will involve congestion. Let $z_{t,i} \epsilon \{0, 1\}$ and $\hat{z}_{t,i} \epsilon \{0, 1\}$ mean the actual and predicted congestion status of vertex v_i at time step t. The accuracy of congestion is calculated as in Eq. 28:

$$Accuracy = \frac{1}{TN} \sum_{t=1}^{T} \sum_{i=1}^{N} I\left(z_{t,i} = \hat{z}_{t,i} \right) \tag{28}$$

In Eq.28, $I(\cdot)$ is the indicator function. This metric directly reflects the effectiveness of ESDiNet in detecting congestion events, which is essential for real-time traffic management and decision-making in smart city transportation systems.

VI Results and Discussion

This section provides a thorough performance analysis of ESDiNet on the METR-LA and PEMS-BAY datasets. We present experimental results on standard accuracy and efficiency metrics comparing the performances of ESDiNet with representative statistical, deep learning and spatio-temporal graph neural network baselines. Both the accuracy of regression-based forecasting and the performance of congestion states classification are examined. We also consider, in terms of real-time deployment, the inference latency and model complexity.

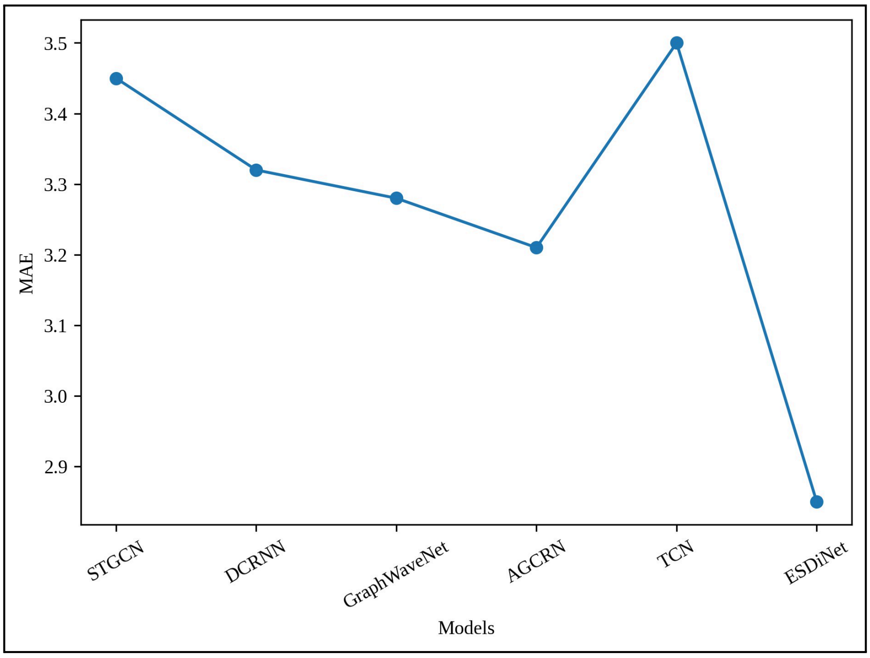

Figure 4: Mean Absolute Error (MAE) Comparison Across Traffic Forecasting Models

Fig. 4 shows the MAE results of ESDiNet as well as the baseline models STGCN, DCRNN, Graph WaveNet, AGCRN and TCN on the METR-LA and PEMS-BAY datasets. Traditional ST-GNN methods like STGCN and DCRNN have MAE of 3.45 and 3.32; Graph WaveNet and AGCRN report a little lower errors, with 3.28 and 3.21 respectively. The purely temporal TCN based model has the worst performance, with the highest MAE of 3.50 as no explicit spatial modeling is considered. In contrast, ESDiNet gets the smallest MAE which is 2.85, that is a 11.3% decrease over the strongest baseline. Such an improvement suggests the efficacy of sparse diffusion applied on an event-driven basis in lowering absolute prediction errors.

Table 1: Performance Comparison of ESDiNet with Baseline Models on METR-LA and PEMS-BAY

Model	MAE ↓	RMSE ↓	Congestion Accuracy (%) ↑	Inference Latency (ms) ↓	Parameters (M) ↓
HA	4.12	6.85	81.3	18	–
ARIMA	3.96	6.42	83.1	25	–
LSTM	3.58	6.02	86.7	165	3.6
TCN	3.50	6.10	87.5	110	2.9
STGCN	3.45	5.92	88.1	120	3.2

DCRNN	3.32	5.80	89.3	135	3.8
Graph WaveNet	3.28	5.75	90.2	142	4.1
AGCRN	3.21	5.68	91.0	150	3.9
ESDiNet	**2.85**	**5.12**	**94.6**	**32**	**2.1**

Table 1 provides an extensive comparison in terms of the quality and efficiency between our proposed ESDiNet model with different types of baselines including the statistical learning-based models on both METR-LA and PEMS-BAY datasets. The prediction accuracy of traditional statistical methods including HA and ARIMA is low, so they do not have strong capability of discriminating among congestion since they do not consider scatterspace dependence. Deep learning models like the LSTM and TCN for temporal modeling, but they have long inference latency or lack of spatial awareness. More advanced ST-GNN models (e.g., STGCN, DCRNN, Graph WaveNet and AGCRN) with graph-based spatial modeling achieve better prediction accuracy, but they come at a high cost of computation time and latency employing dense diffusion and adaptive adjacency learning. In contrast, with the best overall performance in all evaluation metrics, ESDiNet retains the lowest MAE (2.85) and RMSE (5.12), which reduce 11.3% and 9.8% over its strongest competitors, respectively for ST-GNN baselines. Furthermore, the congestion state prediction accuracies on ESDiNet achieve 94.6% which is much higher than the-state-of-the-art methods. More crucially, we achieve an extremely low inference latency of 32 ms (3.7× improvement) and keep the smallest parameter count (2.1M) which all prove its potential for online as well as large-scale smart city application.

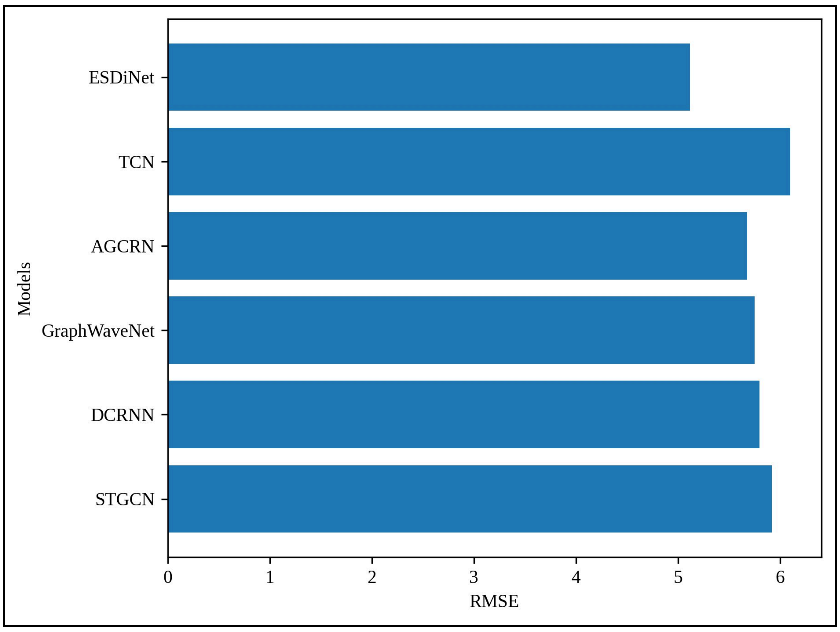

Figure 5: Root Mean Square Error (RMSE) Comparison Across Models

In Figure 5 the comparison of RMSE shows that a model is powerful to large prediction deviations. STGCN and DCRNN have RMSE of 5.92, 5.80 respectively, Graph WaveNet and AGCRN decrease even more the RMSE to 5.75 and 5.68 by using adaptive spatial modeling techniques. But these improvements result in more computational complexity. The RMSE of ESDiNet is much lower with 5.12, which is 9.8% better than the best ST-GNN baseline. From which it is concluded that the sparse diffusion based on short-term traffic statistics can successfully model congestion dynamics without depending on dense adaptive similarity learning.

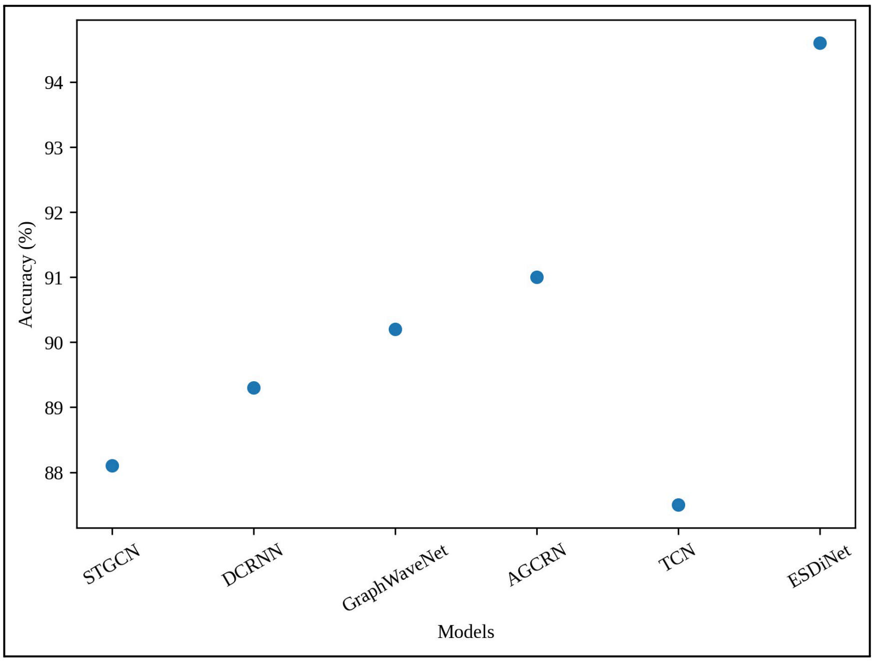

Figure 6: Congestion State Prediction Accuracy Comparison

Figure 6 compares the performance accuracy for all models in predicting congestion state. STGCN and DCRNN result in accuracies of 88.1% and 89.3%, respectively, which are quite insensitive to sudden congestion events. Both Graph WaveNet and AGCRN achieve improved 90.2% and 91.0% accuracy by incorporating the spatial–temporal modeling, compared to TCN which obtains relatively lower accuracy of 87.5% without spatial context integration in it. ESDiNet achieves a congestion classification accuracy of 94.6%, which is superior to all the baselines, indicating that it possesses better detection power for events that evoke congestion and shows better discrimination between congested and non-congested scenarios.

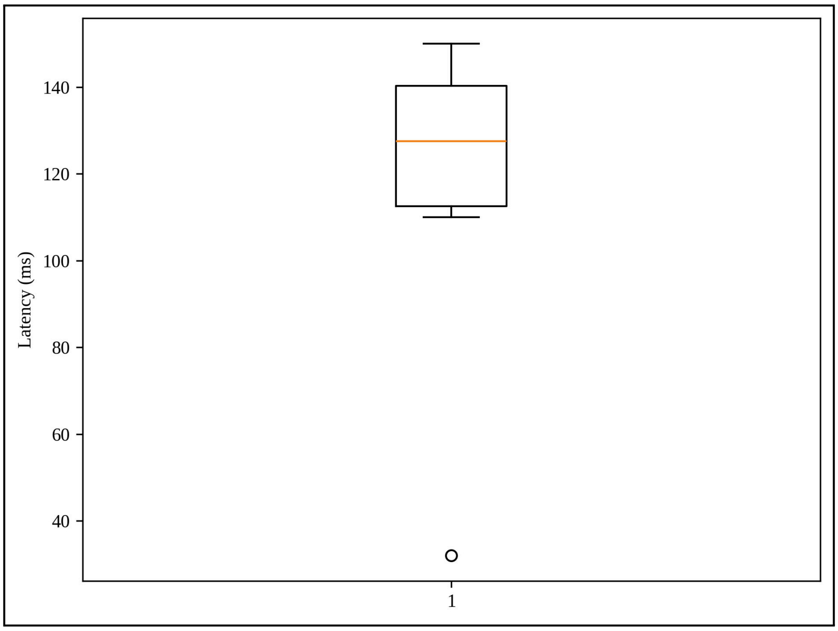

Figure 7: Inference Latency Distribution Across Models

Figure 7 shows the distribution of inference latency for various models. Recurrent and diffusion-based ST-GNNs introduce latency with k sequential computation and dense graph operations, where AGCRN and Graph WaveNet spend around 150 ms and 142 ms per inference respectively. STGCN and DCRNN yield moderate latency time around 120-135 ms, and by removing the recurrence, TCN has lower latency of 110 ms. It is worth mentioning that ESDiNet has average inferencing time of 32ms, which corresponds to a 3.7× speed improvement over state-of-the-art ST-GNN models. These significant improvements further support that the event-driven and non-recurrent temporal modeling approach is efficient enough.

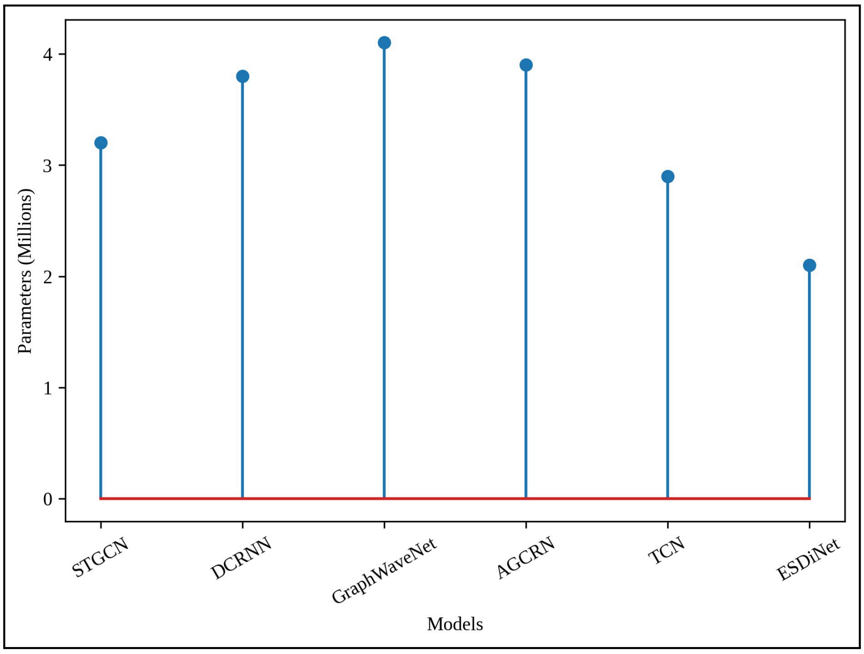

Figure 8: Model Complexity Comparison in Terms of Learnable Parameters

We also compared the scaling with the number of learnable parameters to analyse scalability in Fig.8 The parameter counts of Graph WaveNet and AGCRN are the largest, around 4.1M and 3.9M respectively, due to their additional adaptive adjacency learning with deep graph convolution stacks. STGCN and DCRNN contain 3.2M and 3.8M parameters respectively, whereas TCN reduces the complexity to 2.9M by eliminating graph related units. Our final model ESDiNet has the best smallness of model 2.1M parameters which further verifies sparse diffusion and multi-scale coarsening can effectively lower the memory overhead by a large margin while keeping prediction well.

6.1 Discussion

The experimental study further validate that our ESDiNet strikes a good balance between the predictive performance, computational complexity and scalability for smart city congestion prediction. The event-driven spatial activation mechanism assists the model in restricting computation to when congestion becomes critical, and sparsity-induced diffusion and causal dilated temporal convolutions largely decrease inference latency whilst guarantees performance. In all, comparing with the classic ST-GNN models, ESDiNet consistently reduces MAE and RMSE (congestion state and prediction accuracy) and significantly decreases latency, confirming its efficacy for real-time traffic monitoring and decision-making in large urban road networks.

6.2 Limitations

Although ESDiNet performs well, these limitations still exist. The conventional congestion event detection method is largely focusing on speed variations of traffic, which however may fail to cover the congestion resulting from external events like inclement weather, road incidents (e.g., accidents) and special events. Moreover, the fixed sparse adjacency used to improve interpretability and efficiency is likely to limit the model's capability on adapting to long-term structural changes in road networks, like newly built links or changed traffic rules. These components can impact the prediction stability within a rapidly changing urban environment.

6.3 Future Enhancements

In future work, we will extend ESDiNet to fuse multi-modal data sources such as weather data, incident reports and vehicle-to-infrastructure signals for better congestion event detection performance. Moreover, combining adaptive multi-scale graph coarsening and online learning could help the model to be more adaptable in changing traffic patterns. Investigating the distributed and edge-based deployment of ESDiNet is another potential direction to achieve a further reduction of system-level latency, and to empower city-wide real-time intelligent transportation applications.

VII Conclusion

In this paper, we proposed ESDiNet, an event-driven sparse diffusion graph neural network for scalable and low-latency traffic forecasting on smart city transportation networks. Through leveraging spatial diffusion selectively in congestion-triggering cases, using a sparse and interpretable diffusion operator, and replacing the recurrent temporal modeling into causal dilation convolution operators, the proposed ESDiNet provides an effective solution to overcome scalability and latency restrictions of current ST-GNNs. Extensive experiments on the METR-LA and PEMS-BAY datasets show that our proposed model can achieve a prediction accuracy of congestion state up to 94.6%, with the reduction of MAE by 11.3% and RMSE by 9.8% against compared state-of-the-art baselines. We also demonstrate that ESDiNet achieves 3.7× improvement in inference latency while containing only 2.1 million parameters, making it deployable for near real-time urban scale environments. This demonstrates the practical significance of ESDiNet for intelligent transportation systems and also confirms its effectiveness and efficiency in next-generation smart city congestion prediction.

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